

Crop Classification from Multi-temporal Satellite Images

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Abstract—Accurate crop type classification using remote sensing is critical for agricultural monitoring and food security assessment. This project leverages Google Earth Engine (GEE), Folium, geemap, and TensorFlow to develop and evaluate machine learning models for crop classification using satellite data with multi-temporal dimension. The objective is to compare Long Short-Term Memory (LSTM), Convolutional Neural Network (CNN), Random Forest (RF), and Support Vector Machine (SVM) which are classification models trained on the locational data generated by GEE using spectral and temporal features from Sentinel-2 imagery and USDA NASS Cropland Data Layer (CDL) labels for 2024.

I. AUTHOR CONTRIBUTION

This project task has been assigned to Sezer Enis Birgili and Abdullah Hassoon Younus Alhassoon to work as a group of two. At the early stages as a group, we have chosen the presented idea from a list of available projects presented by the class instructor. Upon withdrawal of Abdullah Hassoon Younus Alhassoon from the class after the add and drop period, expressing his regrets, the study has been carried out by Sezer Enis Birgili alone in all aspects such as, conducting searches, investigating and analyzing data, performing model training and computations, interpreting the results, and writing the final report to present the relevant outcomes all under the close supervision and guidance of Class Instructor Erchan Aptoula.

In this context, it is appropriate to state that the project task was entirely carried out by Sezer Enis Birgili and brought to completion accordingly.

II. INTRODUCTION

In this project, Google Earth Engine, Folium, geemap, and TensorFlow are leveraged to develop and evaluate machine learning models for crop classification using multi-temporal satellite data. The primary objective is to compare multiple classification models such as LSTM, CNN, Random Forest, and SVM, using spectral and temporal features extracted from Sentinel-2 imagery and crop labels from USDA NASS Cropland Data Layer for the year 2024.

Two areas of interest are defined for the data gathering process: an easy region with four dominant crop classes, and a hard region with twenty crop classes. Features including bands B2, B3, B4, and B8, along with Normalized Difference Vegetation Index (NDVI) derived from B4 and B8, are extracted. Cloud coverage is accounted for during data retrieval. The

resulting datasets are exported as CSV files and preprocessed to filter out irrelevant land cover types, removing duplicated data and points with insufficient number of time stamps.

Data is processed in both mono-temporal and multi-temporal formats. In the multi-temporal setting, points with fewer than ten timestamps are excluded, and duplicate spatial-temporal entries are removed. The duplication removal is the same for the mono-temporal format. Summary statistics such as date range, per-point timestamp counts, and observation frequency are also computed.

Visualization includes spatial distribution of samples, class distribution, timestamp statistics, and NDVI time series (for multi-temporal data). Among the models, LSTM achieves the highest absolute accuracy on the hardest classification task once multi-temporal sequences are available, while SVM shows the largest relative gain of any model in moving from mono- to multi-temporal data, on both the easy and hard datasets. RF offers the strongest mono-temporal performance on both datasets, with LSTM and CNN close behind and SVM comparatively the weakest performer in this setting. Across nearly every model and dataset, multi-temporal data vastly improves accuracy over mono-temporal data, the sole exception being RF on the easy dataset, where the gain is marginal.

The code for this implementation of crop classification from multi-temporal satellite images is available on GitHub¹. The data-extraction pipeline, which queries Google Earth Engine directly, is maintained as a separate codebase from the model training and evaluation pipeline, since GEE's free tier imposes strict limits on the number of API calls and re-running extraction on every iteration was not practical.

III. PROBLEM DESCRIPTION

Effective and scalable crop type classification is a critical component of modern agricultural monitoring systems. Accurate identification of crop types over large areas supports a wide range of applications, including yield estimation, land use management, market forecasting, food security analysis, and environmental impact assessment. Traditionally, such classification has relied on manual surveys and local field data, which are often expensive, time-consuming, and difficult to scale. In

¹*Crop Type Classification Using Remote Sensing Machine Learning*. URL: <https://github.com/SezerEnisBirgili/Crop-Type-Classification-Using-Remote-Sensing-Machine-Learning/>.

recent years, remote sensing combined with machine learning has emerged as an alternative, enabling large-scale, automated classification using satellite imagery.

Despite the progress in remote sensing-based classification, several challenges persist. These include high intra-class variability due to different growing stages of the same crop differing in looks from each other (e.g. a growing wheat crop may look brown, green or yellow depending on the growth season), inter-class similarity due to similarities between crops (e.g. wheat and barley grow in similar growth periods and have similar colours), cloud contamination in satellite imagery, and the need for high-quality labeled data. Lastly, using time-based data that shows changes in vegetation, especially for crops with clear growth stages, makes designing and training the model more complicated.

This project addresses the following core research question:

How do different machine learning models perform in classifying crop types using mono-temporal versus multi-temporal satellite imagery, and how does classification complexity affect model accuracy and efficiency?

To explore this question, a pipeline is formulated that uses publicly available satellite data and machine learning techniques to classify crop types in two distinct agricultural regions in the U.S. using CDL as a source of ground truth labels, and Sentinel-2 data to extract spectral features.

Our classification scenarios include:

- **Easy** classification task with four dominant crop classes in eastern Arkansas.
- **Hard** classification task with twenty crop classes in eastern North Carolina.

Each classification task is conducted under two data configurations:

- **Mono-temporal**, where a single observation per point is used.
- **Multi-temporal**, where multiple observations per point are used over the growing season.

Four machine learning models, LSTM, CNN, RF, and SVM, are employed for crop land classification. The performance of each model is compared for each configuration. The aim is to evaluate the impact of temporal data on classification performance, the relationship between classification complexity and model performance and the trade-offs between model accuracy and computational efficiency.

IV. METHODS

This project implements a comprehensive workflow for crop type classification using satellite imagery and machine learning. The process involves several key stages: the defined study areas, data collection, feature extraction, preprocessing, dataset construction, modeling, visualization and evaluation.

A. Study Areas

Two Areas of Interest (AOIs) are defined for the classification tasks:

- **Easy Area of Interest (Eastern Arkansas):** Characterized by a low-diversity agricultural landscape containing

only $n = 4$ distinct crop classes (e.g., predominantly monoculture or simple rotation schemes such as rice, soybeans, cotton, and corn). This region serves as a baseline due to high spectral homogeneity and distinct parcel boundaries.

- **Bounding Box (EPSG:4326):**

[−90.3313°W, 35.5684°N, −90.2865°W, 35.5983°N]

- **Class Complexity:** Low (4 target classes)

- **Hard Area of Interest (Eastern North Carolina):**

Represents a highly complex, heterogeneous agricultural domain comprising $n = 20$ distinct crop classes, including specialty crops and mixed rotations, after non-agricultural and non-crop land cover types were removed (see Section IV-C). The high inter-class spectral overlap and smaller, fragmented field parcels introduce significant classification ambiguity.

- **Bounding Box (EPSG:4326):**

[−77.7215°W, 35.9244°N, −77.5348°W, 36.0472°N]

- **Class Complexity:** High ($n = 20$ target classes)

B. Data Collection

Ground truth labels were extracted from the 2024 USDA NASS Cropland Data Layer, an annually updated, crop-specific land cover dataset for the continental United States. The CDL is generated by the National Agricultural Statistics Service (NASS), Research and Development Division, Geospatial Information Branch, Spatial Analysis Research Section which is available through Google Earth Engine². The CDL provides class labels as numeric codes along with associated color values for visualization. Each code corresponds to a specific crop type, with examples including Corn (1, #ffd400), Cotton (2, #ff2626), Rice (3, #00a9e6), Soybeans (5, #267300), and Winter Wheat (24, #a87000), among many others. Classes considered irrelevant or likely to introduce noise—such as open water, urban/developed areas, barren land, natural vegetation, or non-active cropland, were excluded from sampling; the full rationale for these exclusions is given in Section IV-C. The numerical CDL codes were mapped to human-readable crop names for better readability during pre-processing. This mapping also included the standard CDL color codes to enable consistent labeling in visualization outputs, such as maps and class distribution charts.

C. Excluded Land Cover and Crop Classes

Following the initial extraction of CDL-labeled points, several categories of land cover and crop labels were removed prior to model training. These exclusions were applied consistently to both AOIs and are the reason the Hard AOI is reduced to $n = 20$ crop classes rather than the full set of CDL codes present in the raw extraction. The excluded categories fall into four functional groups.

Developed land (all intensity levels), developed open space, open water, and barren land were removed. These classes

²USDA NASS Cropland Data Layers, Google Earth Engine Data Catalog. URL: https://developers.google.com/earth-engine/datasets/catalog/USDA_NASS_CDL.

contain no vegetation signal, or in the case of developed open space and barren land, contain signals dominated by managed turf, impervious surfaces, or bare soil rather than commercial crops. Including them would introduce spectral noise unrelated to crop phenology and risk false positives in a model intended specifically for crop classification.

Deciduous, evergreen, and mixed forest, shrubland, woody wetlands, and grass/pasture were removed. Although these classes are vegetated and exhibit seasonal greenness, their growth patterns do not follow the planting, peak-greenness, and harvest cycle characteristic of managed row crops. Evergreen canopy is largely static across seasons, deciduous leaf-on/leaf-off timing does not align with crop calendars, and pasture is either grazed irregularly or left uncultivated, none of which produce a phenology curve comparable to a cultivated crop.

Fallow/idle cropland and Other Hay/Non-Alfalfa were removed. Fallow land presents as bare soil, weeds, or crop residue and carries no active crop signal to classify. Other Hay/Non-Alfalfa is a catch-all category covering wild hay and mixed-species forage grasses with highly variable spectral signatures and irregular, unmanaged harvest schedules, unlike a specifically cultivated forage crop such as Alfalfa.

All Double Crop labels (e.g., winter wheat/soybeans, corn/soybeans) were removed. These fields undergo two distinct planting-to-harvest cycles within a single calendar year, producing a bimodal temporal signature that does not correspond to any single target crop class. They also make up a small fraction of the labeled points relative to single-crop fields, so retaining them would introduce both label ambiguity and class imbalance without a meaningful gain in usable samples.

D. Feature Extraction

Spectral and temporal features were extracted from Copernicus Sentinel-2 Harmonized Surface Reflectance (H-SR) imagery³. Sentinel-2 is a high-resolution, multi-spectral imaging mission developed by the European Space Agency (ESA) as part of the Copernicus Program. It supports applications in agriculture, forestry, water resources, and land monitoring. From each image, The following Level-2A spectral bands were extracted:

- **B2 (Blue)** — 10m resolution, 492–497 nm
- **B3 (Green)** — 10m resolution, 559–560 nm
- **B4 (Red)** — 10m resolution, 664–665 nm
- **B8 (Near Infrared, NIR)** — 10m resolution, 833–835 nm

Using the red (B4) and NIR (B8) bands, The Normalized Difference Vegetation Index (NDVI) at each timestamp is computed as:

$$\text{NDVI} = \frac{B8 - B4}{B8 + B4}$$

³Harmonized Sentinel-2 MSI: MultiSpectral Instrument, Level-1C (TOA), Google Earth Engine Data Catalog. URL: https://developers.google.com/earth-engine/datasets/catalog/COPERNICUS_S2_HARMONIZED.

NDVI shows how healthy and active vegetation is over time and was central to the temporal modeling strategies applied.

To ensure data quality, only cloud-free observations were retained. GEE’s built-in cloud masking approach was applied using the QA60 band. Future studies may incorporate Red Edge or SWIR bands to enrich the spectral feature space, particularly for distinguishing subtle crop differences or stress responses.

E. Preprocessing I& Dataset Construction

Two dataset formats were prepared to be applied to both AOIs:

Experimental Observational Setups

- 1) **Mono-temporal Setup:** Captures a single peak-season spectral observation per location to evaluate classification performance. Here, one “sample” corresponds directly to one sampled location, and a minimum of 50 locations per class was required. For every location, the single retained observation was the one closest to July 15, 2024 (mid-growing season), used as a common target date across all points.
- 2) **Multi-temporal Setup:** Incorporates full spectral time-series data points collected across the entire growing season to use phenological changes in crops. Here, one “sample” corresponds to one timestamp record rather than one location; a minimum of 1,000 timestamps per class was required, which corresponds to roughly 50 locations per class at an average of 20 timestamps per location.

Dataset Construction & Preprocessing Pipeline

First, **deduplication** removed duplicate records sharing identical coordinate and temporal keys (lon, lat, date); a point was treated as a duplicate only if both its location and its observation date matched an existing record, and the count of such duplicates was checked directly before removal. Next, **temporal quality filtering** excluded sparse time-series observations by discarding points with fewer than 10 valid timestamps across the monitoring window. Following this, **label mapping and interpretability** was addressed: the original ground truth labels obtained from the Cropland Data Layer (CDL) were encoded as integer class IDs, and these numeric codes were then mapped to human-readable crop names using a predefined lookup dictionary for better interpretability. In addition, **agricultural class filtering** was applied to retain only agriculturally relevant land cover classes, as detailed in Section IV-C; this step avoided learning really small classes and ensured that the models focused solely on crop-type classification rather than broader land use classification. Beyond this, **feature normalization and scaling** was performed to standardize the feature scales across inputs and reduce the risk of model bias toward high-magnitude features, so spectral bands (B2, B3, B4, B8) and NDVI values were normalized; this is particularly important for neural network models, which are sensitive to feature scale and distribution.

Finally, **temporal alignment and padding** was carried out because satellite imagery is collected on irregular dates due to cloud cover, satellite orbit, and data availability, meaning the resulting time series varied in length across locations; to enable batch processing and compatibility with fixed-length sequence models, all time series were therefore aligned to a common temporal window, which was achieved by padding to the maximum amount of timestamps in a given dataset.

The cleaned and preprocessed datasets were exported as comma-separated values (.csv) files. Specifically, observations corresponding to the low-complexity region were saved to `crops_easy.csv`, while data for the high-diversity region were exported to `crops_hard.csv`.

F. Machine Learning Models

Four machine learning models were evaluated for crop classification across both the easy and hard datasets, under mono-temporal and multi-temporal configurations. Each model was selected for its unique strength in handling different data modalities:

1) Long Short-Term Memory (LSTM):

LSTM networks are a type of recurrent neural network (RNN) specifically designed to capture long-term dependencies in sequential data. In our pipeline, LSTM was applied to both mono-temporal and multi-temporal datasets to leverage the phenological patterns captured across time. Although in the case of mono-temporal it was heavily modified to accept non-sequential data as such the performance was not good. At a high level, the network masks padded timesteps, passes the sequence through a bidirectional LSTM layer with layer normalization and dropout, applies multi-head self-attention so the model can weight the most informative timesteps, pools the resulting sequence, and finishes with a dense block (batch normalization, GELU activation, dropout) before the final softmax classification layer.

2) Convolutional Neural Network (CNN):

CNNs are effective at extracting local patterns in structured input such as images or time series. For mono-temporal data, a simple 1D CNN was applied directly to feature vectors. For multi-temporal data, time series were reshaped into a 2D format allowing the model to apply 2D convolutions across both time and spectral dimensions. This made CNNs capable of learning both temporal trends and inter-band relationships. At a high level, the network masks padded timesteps, passes the sequence through two 1D convolutional blocks (64 and 128 filters) each with batch normalization, GELU activation, and dropout, combines global average and global max pooling to summarize the sequence, and finishes with a dense block (batch normalization, GELU activation, dropout) before the final softmax classification layer.

3) Random Forest (RF):

Random Forest is an ensemble method based on decision

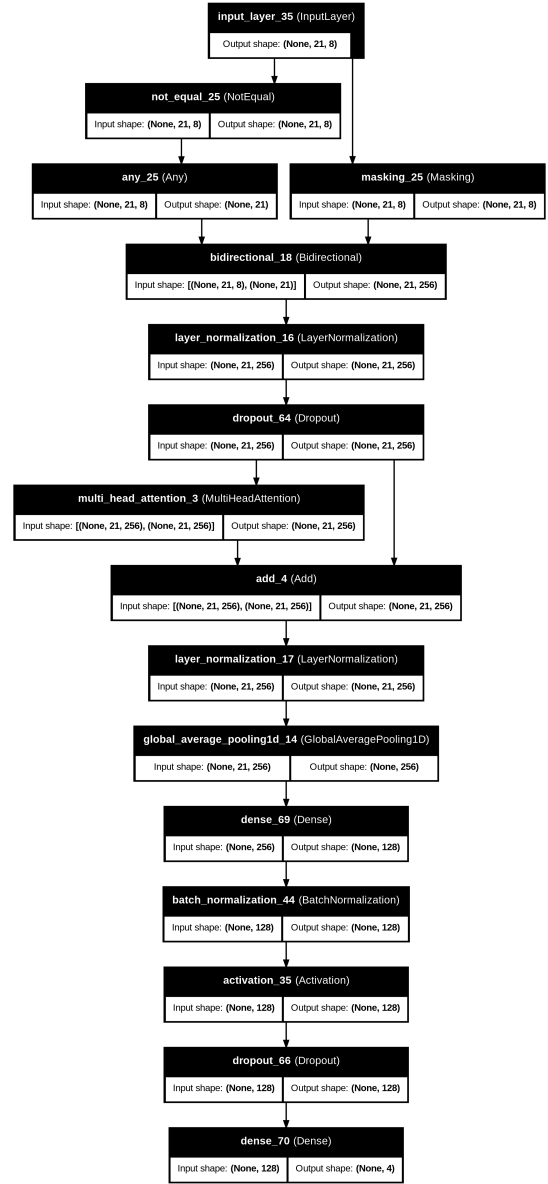


Fig. 1. LSTM architecture used for multi-temporal crop classification. Input features per timestep included NDVI and four spectral bands.

trees. It constructs multiple trees during training and outputs the mode of the classes for classification tasks. RFs are robust to overfitting, easy to implement, and require minimal hyperparameter tuning. They were particularly effective for mono-temporal classification due to their ability to handle high-dimensional tabular data.

4) Support Vector Machine (SVM):

SVMs find the optimal hyperplane that separates classes in high-dimensional space. The radial basis function kernel, which is well-suited for nonlinear classification tasks was used for the SVM model. While SVMs performed well on simpler, mono-temporal, low-class-count, datasets their scalability limited their utility on multi-temporal or high-class-count datasets.

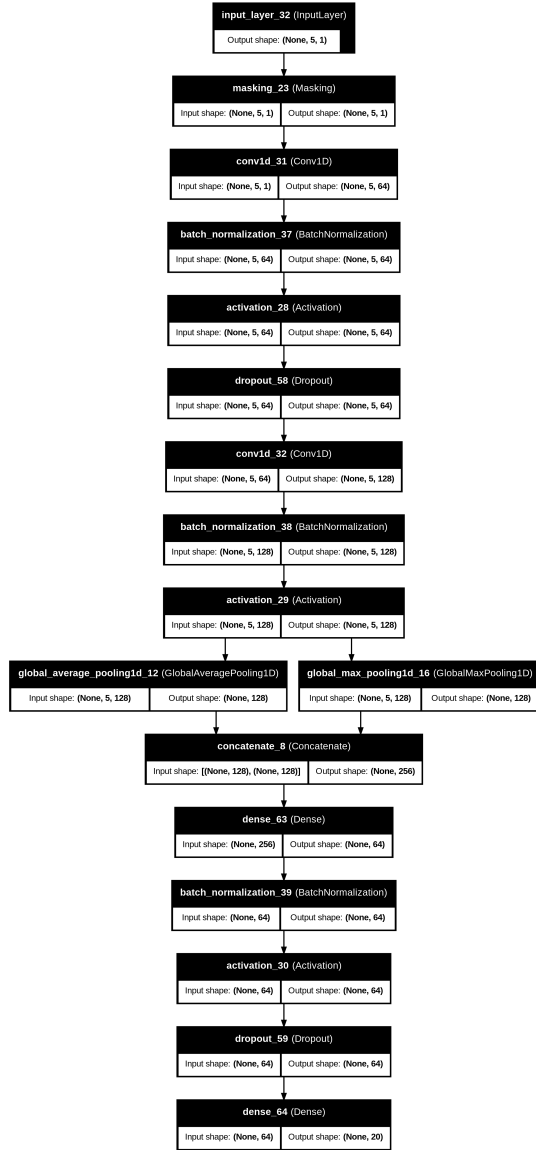


Fig. 2. CNN architecture for time-feature grid input. Applied to both mono- and multi-temporal data using 1D or 2D convolutions.

Models were implemented using TensorFlow and Scikit-learn. An 80/20 train-test split was used. Both the LSTM and CNN models were trained with a batch size of 32 for up to 80 epochs, although training typically stopped after around 20 epochs once validation performance plateaued.

G. Visualization

Folium and geopmap were used to create interactive maps and exploratory visualizations. Visual diagnostics were essential for understanding dataset characteristics and interpreting model behavior. Below are examples of key visualizations for both mono- and multi-temporal dataset configurations for both easy and hard datasets.

1. Spatial Distribution of Labeled Points with NDVI Basemaps: Crops are evenly distributed within the given AOI boundaries in both datasets, as shown in Figure 3. However,

both the easy and hard regions demonstrate class imbalance, where certain crops dominate the label distribution. The NDVI basemaps further confirm that labeled points span various vegetation zones, providing representative coverage of land cover conditions.



Fig. 3. Spatial distribution of labeled points over NDVI basemaps.

2. Spatial Distribution of Labeled Points with Respect to Class: As visualized in Figure 4, class-specific spatial distributions reveal how certain crop types cluster geographically. While crops are generally dispersed across the AOI, dominant classes such as corn or soybean tend to concentrate in specific subregions. This spatial bias may impact model generalization if some crops are limited to narrow zones.

3. Class Distribution Histograms: Figure 5 shows the frequency of labeled samples for each class. Both easy and hard datasets exhibit significant class imbalance, where crops like corn and soybean dominate while classes such as barley and oats are underrepresented. This imbalance is more pronounced in the hard dataset and directly affects classification performance on minority classes.

4. Distribution of Timestamps (Only Multi-Temporal): The distribution of valid timestamps across samples is illustrated in Figure 6. Both datasets cover observations throughout the growing season, providing sufficient temporal coverage to detect phenological patterns. Most samples include at least 10 timestamps, satisfying the minimum threshold for LSTM-based models. While the 10-timestamp minimum admitted some sparser points, the majority of retained points clustered around the dataset average of roughly 20 timestamps. Overall seasonal progression is well-represented, with observations spanning from March 10, 2024 to October 21, 2024. The interval between consecutive valid observations at a given location varied moderately rather than sharply: gaps of 5, 10, and 15 days were most common, consistent with Sentinel-2's revisit cycle, with occasional longer gaps of up to 20-25 days where persistent cloud cover excluded intermediate acquisitions.

5. NDVI Time Series by Crop Type: A core premise of the multi-temporal approach is that distinct crop types follow distinguishable growth trajectories over the season, and that this pattern is learnable from NDVI alone. To confirm this was possible before modeling, NDVI time series were plotted for representative points from the most common crop classes in each AOI: five crop types for the hard dataset and four for the easy dataset, shown in Figure 7. The resulting curves

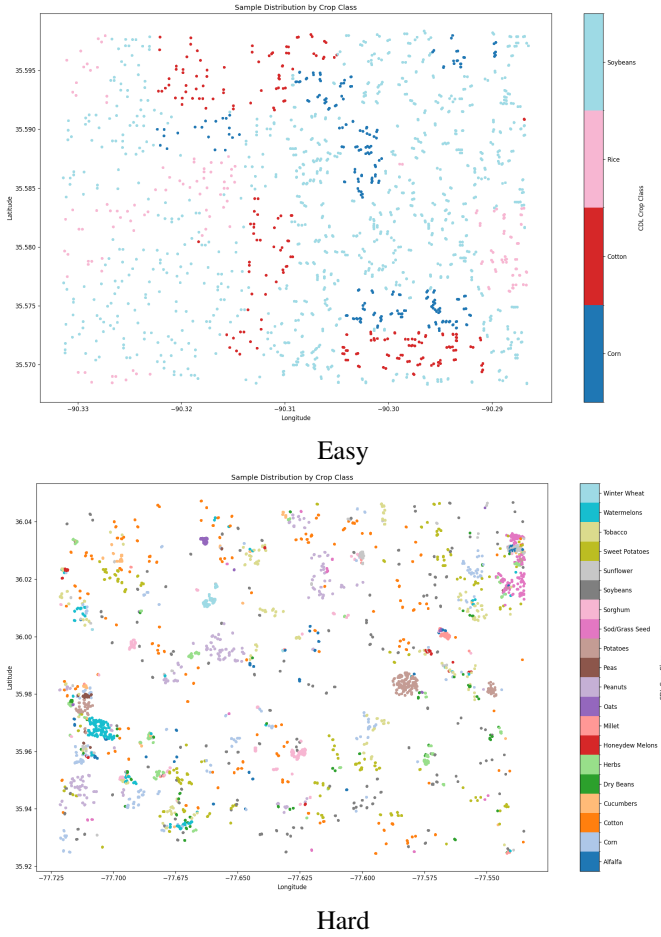


Fig. 4. Spatial distribution of labeled points by class.

show visibly different NDVI trajectories across crop types, supporting the use of temporal sequences as a discriminative signal for classification.

V. RESULTS

To evaluate the effectiveness of our classification pipeline, a series of experiments were conducted across the aforementioned two locations (easy and hard) with two data types (mono-temporal and multi-temporal). Each of the four configurations were used to train and test four models: LSTM, CNN, RF, and SVM. The results below summarize classification performance, temporal effects, and class-level behavior.

A. Experiment Objectives

Each experiment aimed to answer the following research questions:

- 1) How does classification accuracy vary between mono-temporal and multi-temporal datasets?
- 2) Which model performs best in each setting (easy vs. hard, mono vs. multi)?
- 3) What is the impact of temporal information on class separability?
- 4) How does model performance trade off with computational cost?

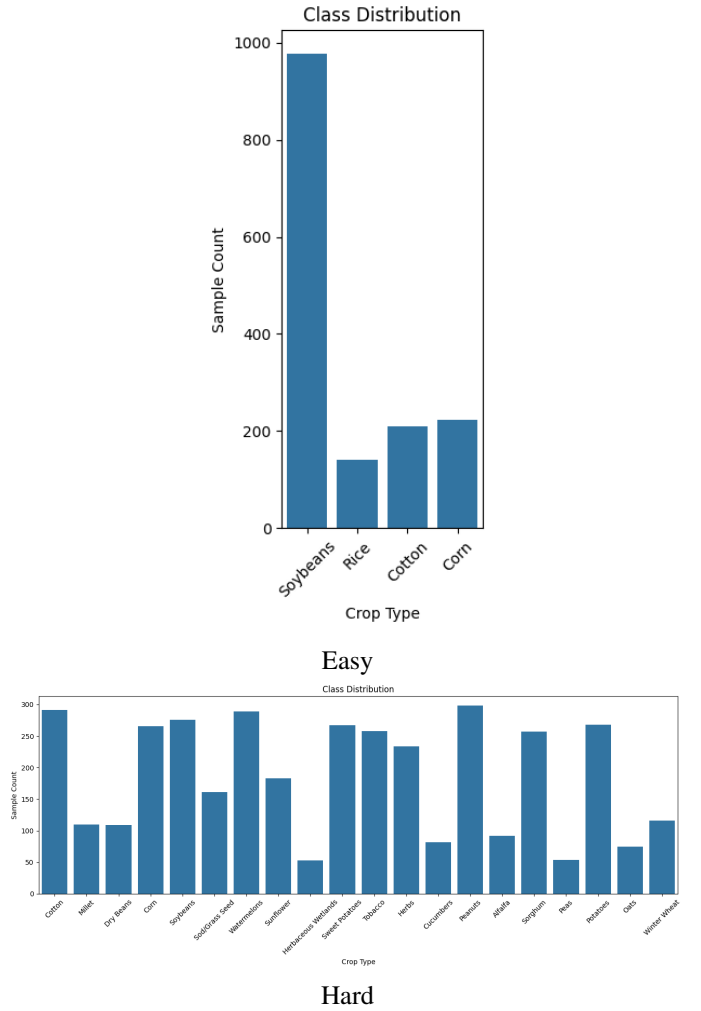


Fig. 5. Class distribution histograms across dataset configurations.

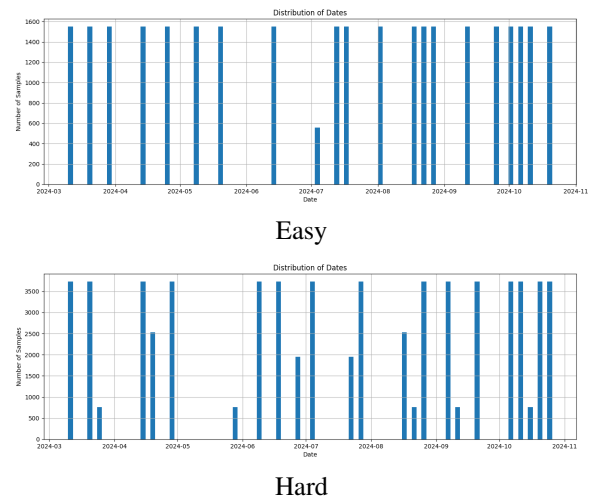


Fig. 6. Distribution of timestamps per point in multi-temporal datasets.

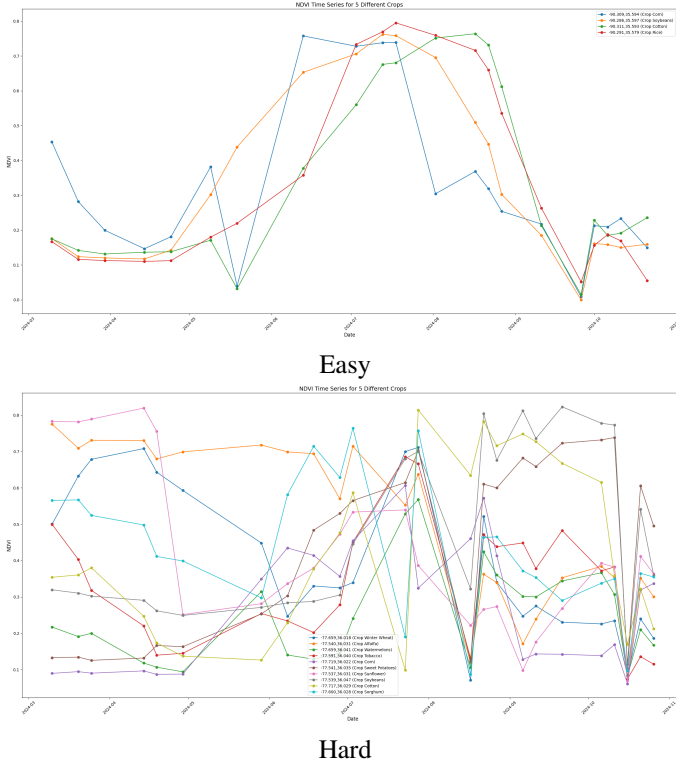


Fig. 7. NDVI time series for representative points across different crop types, illustrating distinct growth trajectories over the monitoring window.

B. Overall Performance

Table I summarizes weighted F1-scores across all 16 experiments:

TABLE I
WEIGHTED F1-SCORES BY MODEL, DATASET, AND TEMPORAL CONFIGURATION.

Model	Easy Mono	Easy Multi	Hard Mono	Hard Multi
LSTM	0.79	0.96	0.21	0.79
CNN	0.76	0.95	0.20	0.77
RF	0.88	0.96	0.31	0.76
SVM	0.66	0.96	0.14	0.75

Key findings:

- Multi-temporal data vastly improved accuracy over mono-temporal data across every model, with a single exception: RF on the easy dataset, where the gain was marginal (0.88 to 0.96, +0.08) compared to double-digit-point gains everywhere else.
- **LSTM** achieved the highest absolute score on the hardest task (Hard Multi, 0.79), improving by 0.58 over its own mono-temporal score. However, **SVM** posted the largest relative gain of any model on both datasets (+0.61 on Hard, from 0.14 to 0.75; +0.30 on Easy, from 0.66 to 0.96), despite still trailing LSTM in absolute Hard Multi performance.
- **RF** performed best on mono-temporal setups on both datasets, with **SVM** consistently the weakest performer

in this setting (RF 0.88 vs. SVM 0.66 on Easy Mono; RF 0.31 vs. SVM 0.14 on Hard Mono).

- **CNN** provided balanced and stable results across all cases, generally following just behind LSTM.
- On multi-temporal data, RF and SVM scores converge and become comparable to LSTM and CNN. This convergence does not hold in the mono-temporal setting, where SVM lags well behind the other three models.

C. Confusion Matrices for Best Models

Below are the confusion matrices for the best-performing model in each case:

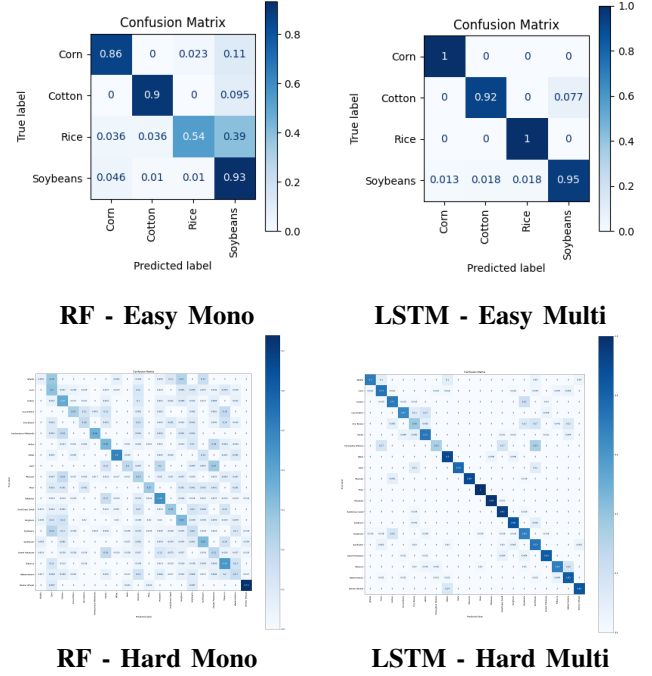


Fig. 8. Confusion matrices for best models across dataset configurations.

D. Class-Level Accuracy Histograms

The following histograms illustrate per class accuracy for the same four best-performing models:

E. Impact of Data Characteristics

- Strong class imbalance in the hard datasets reduced model generalizability.
- The minimum threshold of 10 valid timestamps effectively filtered low-quality samples for LSTM.
- Temporal signals in NDVI notably improved separation of crops with similar reflectance patterns.

VI. DISCUSSION

The experimental results provide several important insights into the effectiveness of different machine learning approaches for crop classification using remote sensing data. The comparison across datasets of varying complexity and temporal configurations reveals key trends in model performance, data utility, and computational trade-offs.

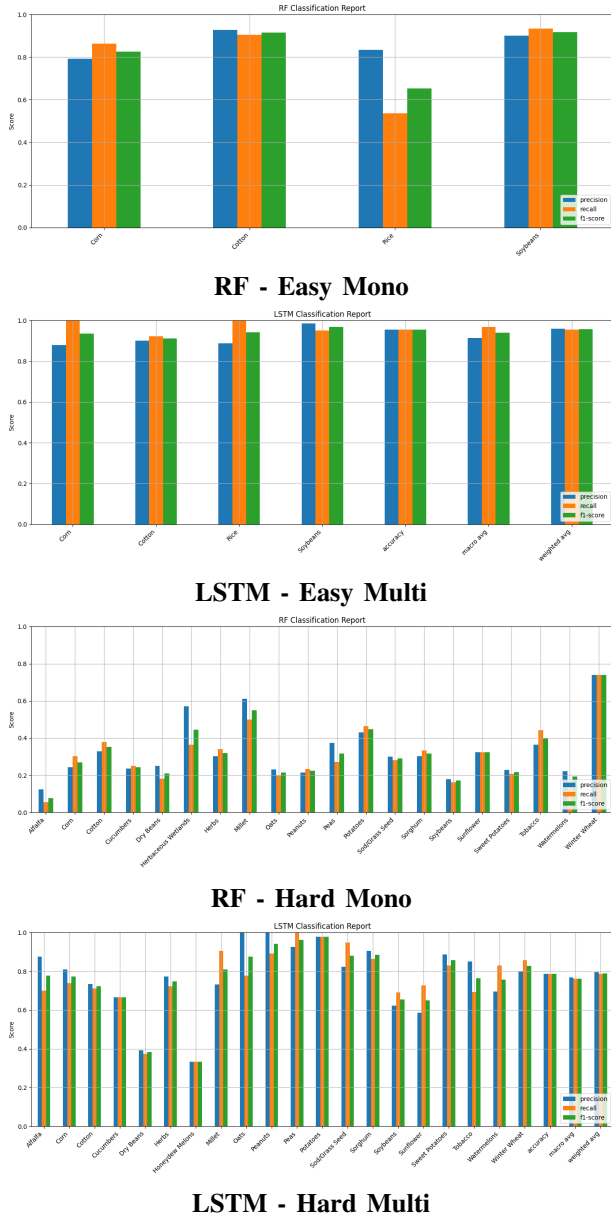


Fig. 9. Per-class accuracy histograms for top-performing models.

A. Impact of Temporal Information

One of the most significant findings is that multi-temporal data vastly improves classification accuracy over mono-temporal data, and this holds regardless of model, with a single exception: RF on the easy dataset, where the improvement was smaller (+0.08) compared to the massive double-digit-point gains seen everywhere else. This confirms that temporal information is broadly and consistently valuable across model architectures, not just for any one method. In relative terms, SVM benefited the most from the addition of temporal information, gaining +0.61 on the Hard dataset (0.14 to 0.75) and +0.30 on the Easy dataset (0.66 to 0.96), the largest improvements of any model on either dataset. LSTM’s own gain on the Hard dataset was nearly as large (+0.58, from 0.21 to 0.79)

and, combined with its already competitive mono-temporal accuracy, made it the top absolute performer on the hardest classification task once multi-temporal data was available, edging out CNN (0.77), RF (0.76), and SVM (0.75) on Hard Multi. This is likely due to its ability to capture phenological patterns over time, which are crucial for distinguishing crops with similar spectral signatures but differing growth cycles. SVM’s larger relative jump, by contrast, is best read against its much weaker mono-temporal starting point (the lowest of the four models on both datasets) rather than as evidence that it becomes the strongest multi-temporal model.

CNNs also benefited from temporal data when reshaped into 2D structures (e.g., time versus features), though not to the same extent as LSTMs. Traditional models like RF and SVM achieved high baseline performance on mono-temporal data in RF’s case, but both still scaled significantly when the temporal dimension was added. In the case of “Hard Multi”, their scores improved to be highly comparable to the LSTM and CNN. This is likely due to the fact that with more classes there is a need for more data to differentiate between crop types, as mono-temporal data might not be enough for twenty classes of crop types.

B. Effect of Dataset Complexity

Model performance declined across all methods as dataset complexity increased. In the hard dataset, which included 20 crop types, overall accuracy were lower than in the easy dataset (with four dominant classes). This can be attributed to:

- Increased class imbalance, leading to overfitting on dominant crops.
- Greater spectral and temporal overlap between crop types.
- Limited training samples per class, even with balanced sampling efforts.

While LSTM still achieved the highest performance on the hard multi-temporal dataset, confusion matrices revealed difficulties in distinguishing minority classes. This highlights the need for additional training and data to differentiate between the crop types.

C. Trade-offs Between Accuracy and Efficiency

This study did not include formal training-time benchmarking, so the following is a qualitative characterization rather than a measured result. RF and SVM are, by construction, computationally lighter than the neural models: neither requires iterative gradient-based optimization over many epochs, and RF in particular is known for fast training and minimal hyperparameter tuning. These properties make them attractive for quick baselines or deployments in resource-limited settings, and RF in particular is also the strongest performer of the four models on mono-temporal data. However, RF and SVM offer more limited performance gains on multi-temporal data, especially on the hard, high-class-count dataset.

In contrast, the neural models (CNN and LSTM) require iterative training with a batch size of 32 for up to 80 epochs, and generally more tuning, but achieve higher accuracy on the more difficult, multi-temporal task. These trade-offs suggest

that model selection should be done on a case-by-case basis where cost, speed, and accuracy are assessed together to fulfill a specific goal; a direct empirical comparison of training and inference time across models is left to future work.

D. Limitations

While promising, the current study has several limitations:

- The use of a minimum of 50 sampled locations per class (corresponding to roughly 1,000 timestamp records per class in the multi-temporal setting) may be insufficient for rare crops, especially in high-diversity datasets.
- Cloud masking, while robust, may still admit artifacts that degrade temporal signal quality.
- Mono-temporal data relied on a single mid-season date, which may not represent all crop phenologies equally.
- Only two AOIs were selected, so the results cannot be generalized to a given crop class everywhere in the world; the same crop grown in a different climate or region may exhibit different spectral behavior or growth periods than what was observed here.
- The hard dataset contains roughly 168,000 timestamps across about 8,700 points, while the easy dataset contains roughly 32,000 timestamps across about 1,500 points. More data may be needed to improve performance; the easy dataset's small size relative to its low class count is suspected to encourage overfitting, while the hard dataset's larger size still appears insufficient for its 20 classes, suggesting it requires substantially more data or more robust models to reach comparable accuracy.
- No training or inference time measurements were collected in this study; claims about relative computational efficiency across models are therefore qualitative and should be validated empirically in future work.

Future work could explore enhanced sampling strategies, working with more data to search the limit for accuracy for the current configurations, and using more advanced sequence models such as Transformers to better leverage temporal dependencies.

VII. CONCLUSIONS

This study demonstrates the feasibility of combining remote sensing and machine learning techniques for crop type classification across spatial and temporal variations. Leveraging Google Earth Engine, Sentinel-2 imagery, and USDA Cropland Data Layer labels, a reproducible pipeline capable of classifying crops in two distinct agricultural regions using both mono-temporal and multi-temporal datasets was developed and evaluated.

Our results highlight the significant benefits of temporal information: multi-temporal data vastly outperformed mono-temporal data across virtually every model tested, the lone exception being RF on the easy dataset. Models such as LSTM, which are designed to capture sequential patterns, achieved the highest classification accuracy on the hardest multi-temporal task, and were most valuable precisely on the hardest, 20-class task, where the added temporal signal was

needed most; on the simpler easy-region task, multi-temporal accuracy converged to a similar, near-ceiling score across all four models. Among the four evaluated models: LSTM, CNN, RF, and SVM, LSTM achieved the strongest performance in the more demanding multi-temporal, high-class-count scenario (Hard Multi), while RF was the strongest performer on mono-temporal data and, along with SVM, is architecturally simpler and less computationally demanding to train, making these two models attractive where speed and simplicity are prioritized over squeezing out the last few points of accuracy on complex, multi-temporal tasks.

It was also found that dataset complexity substantially influenced model accuracy. Careful dataset construction and preprocessing, including filtering and relabeling, proved critical to success. Visualizations such as NDVI time series and spatial class distributions were quite useful in diagnosing model behavior and informing methodological decisions.

Future directions for this research include:

- Exploring enhanced sampling strategies to improve data quality and coverage.
- Using larger datasets to test the accuracy limits of the current model configurations.
- Investigating advanced sequence models, such as Transformers, to better capture temporal dependencies.
- Collecting empirical training and inference time measurements to substantiate computational-efficiency comparisons across models.

REFERENCES

- [1] *Crop Type Classification Using Remote Sensing Machine Learning*. URL: <https://github.com/SezerEnisBirgili/Crop-Type-Classification-Using-Remote-Sensing-Machine-Learning/>.
- [2] *Harmonized Sentinel-2 MSI: MultiSpectral Instrument, Level-1C (TOA), Google Earth Engine Data Catalog*. URL: https://developers.google.com/earth-engine/datasets/catalog/COPERNICUS_S2_HARMONIZED.
- [3] *USDA NASS Cropland Data Layers, Google Earth Engine Data Catalog*. URL: https://developers.google.com/earth-engine/datasets/catalog/USDA_NASS_CDL.

APPENDIX

Additional Results

Additional results have been provided herein after in the next pages as graphs (see Figures 10, 11, 12, 13, 14, 15) to elaborate on the accuracy levels of crop classification for other configurations and models broken up according to class.

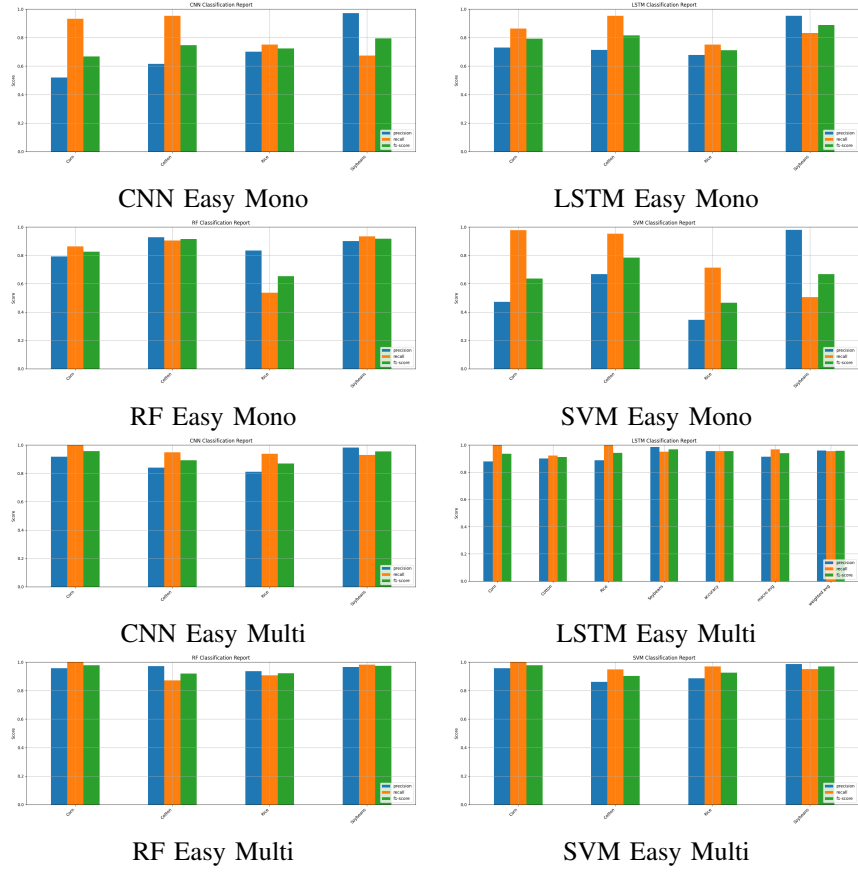


Fig. 10. Per-class accuracy metrics (precision, recall, F1) across all dataset configurations and models.

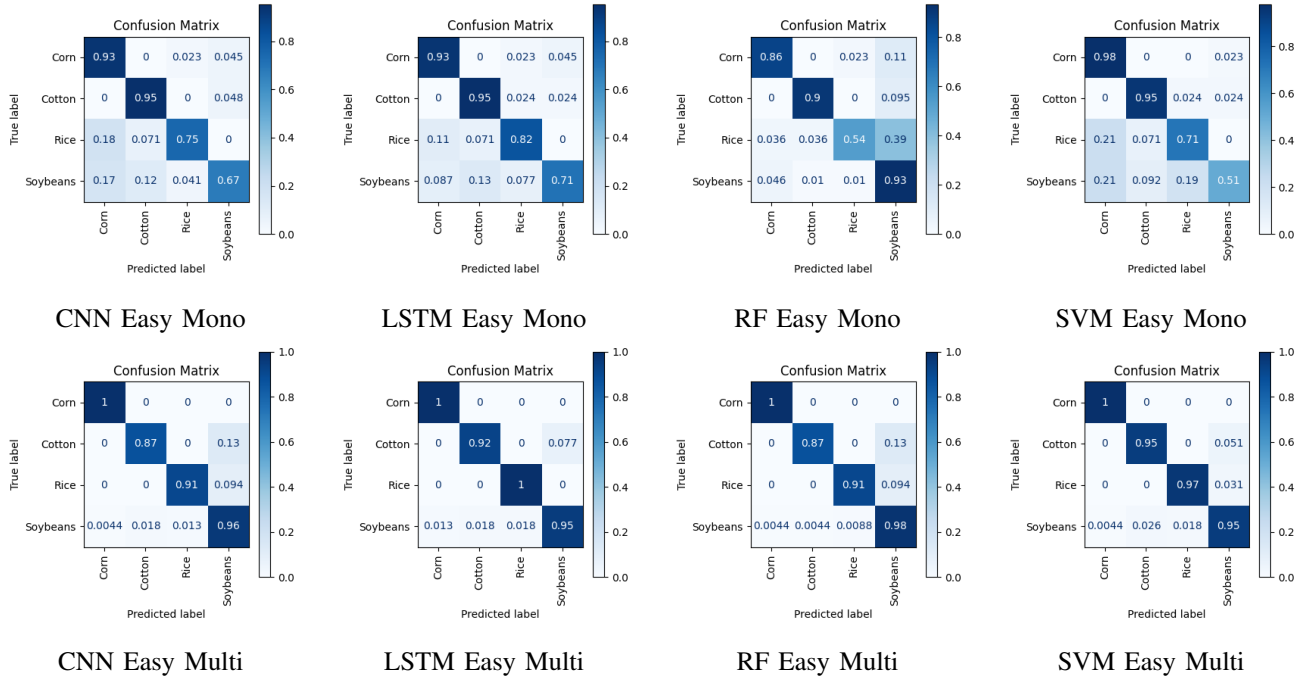
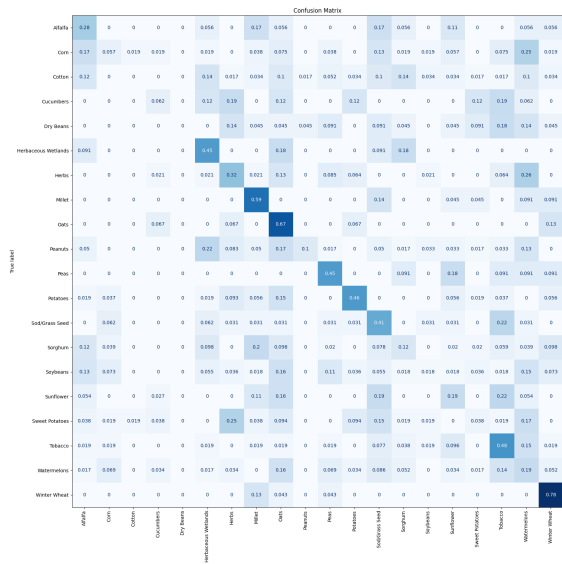
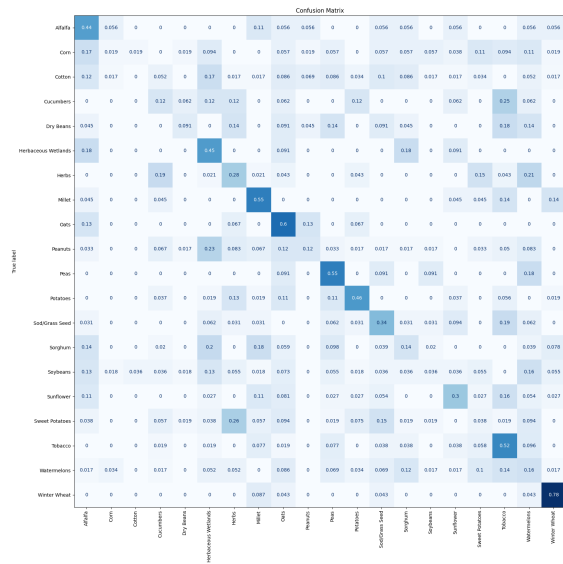


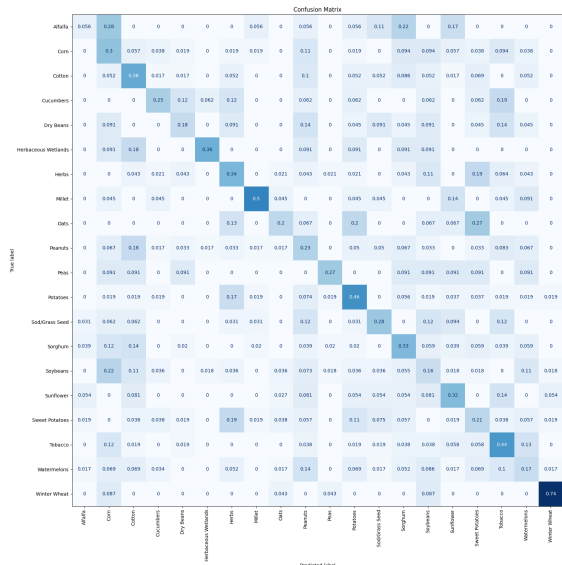
Fig. 11. Confusion matrices for each model on the easy dataset.



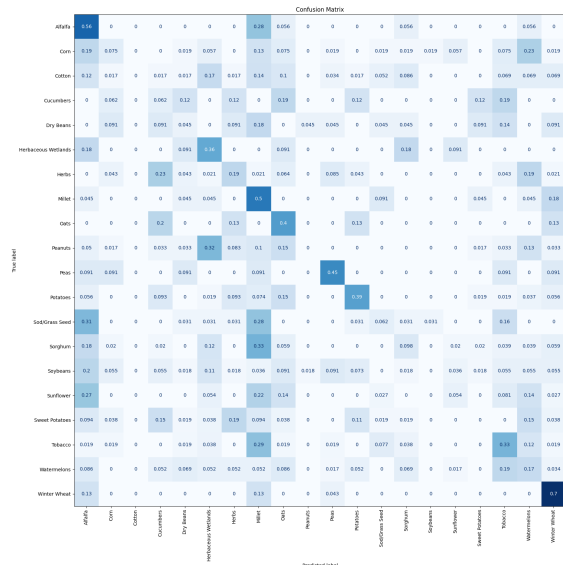
CNN Hard Mono



LSTM Hard Mono

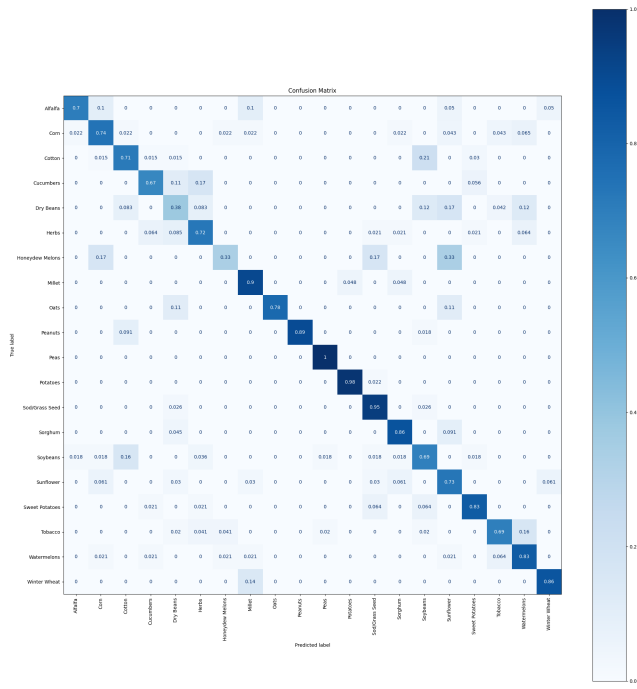


RF Hard Mono

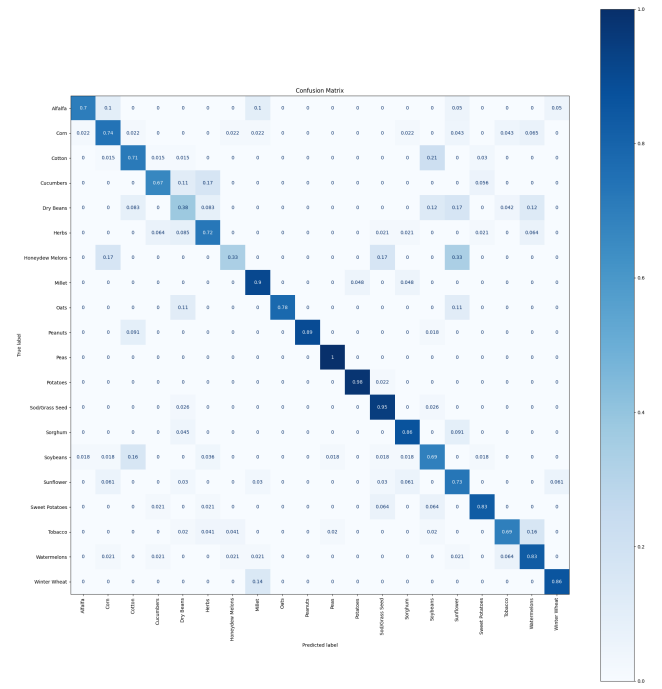


SVM Hard Mono

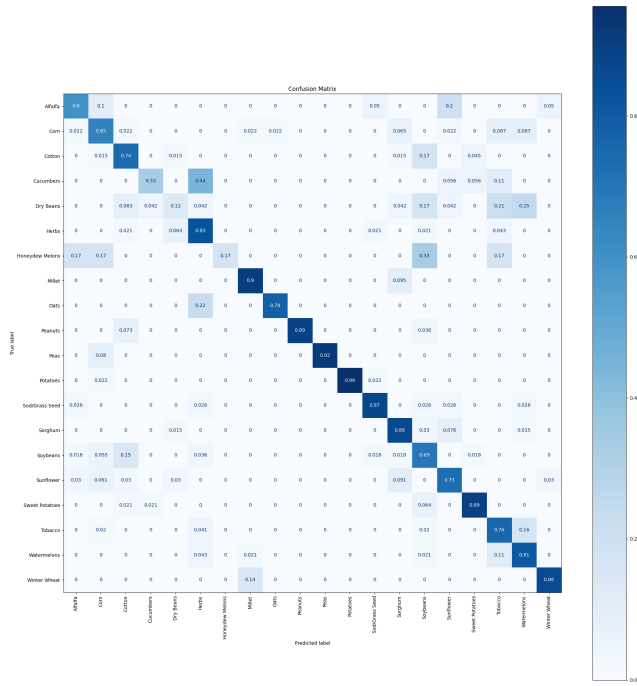
Fig. 12. Confusion matrices for each model on the Hard Mono dataset.



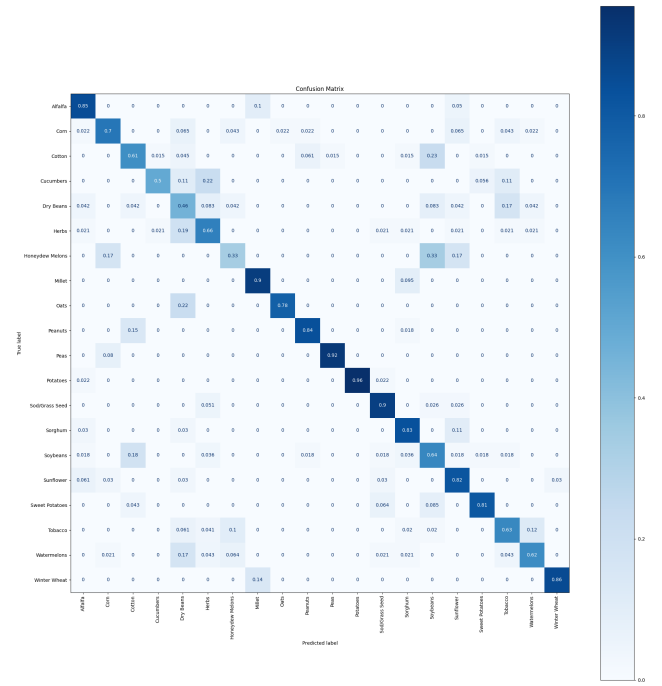
CNN Hard Multi



LSTM Hard Multi

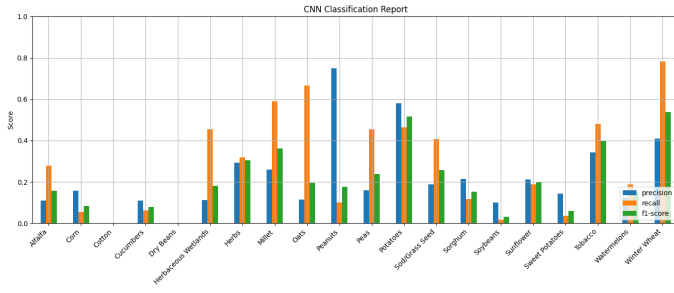


RF Hard Multi

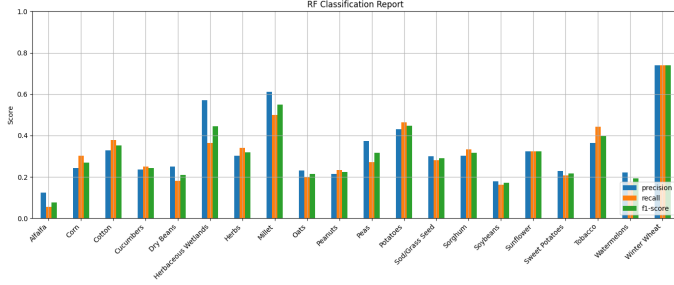


SVM Hard Multi

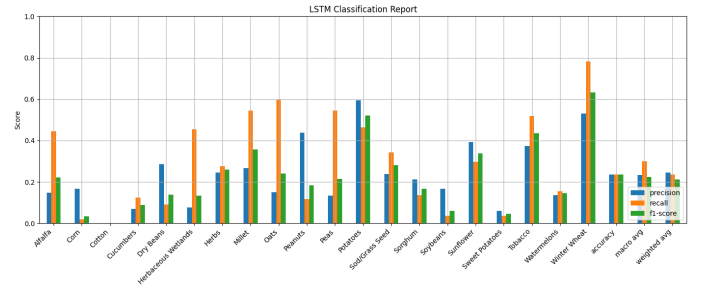
Fig. 13. Confusion matrices for each model on the Hard Multi dataset.



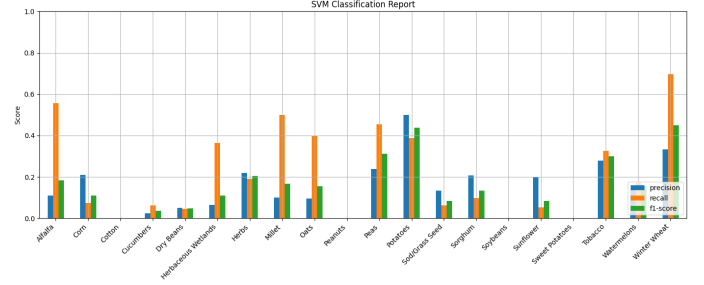
CNN Hard Mono



RF Hard Mono

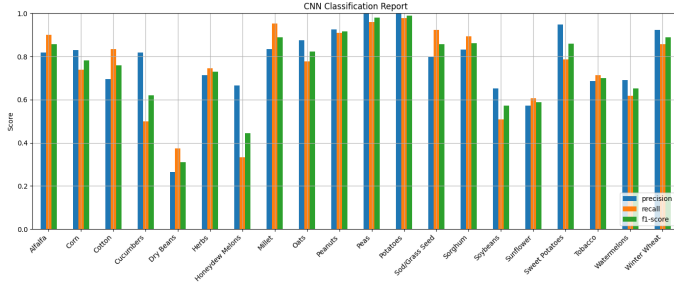


LSTM Hard Mono

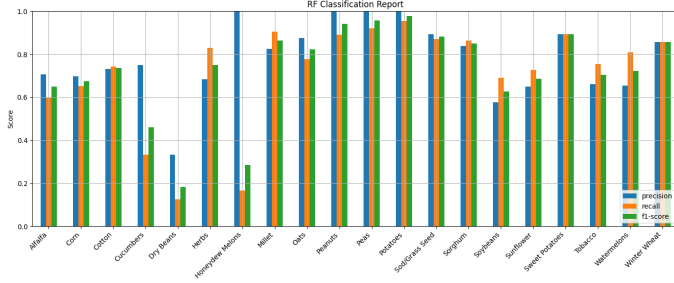


SVM Hard Mono

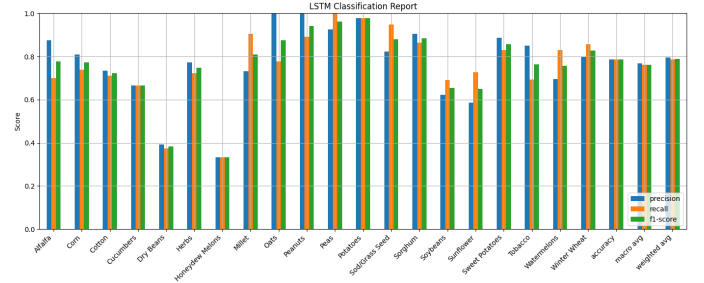
Fig. 14. Per-class accuracy metrics (precision, recall, F1) for the Hard Mono dataset across all models.



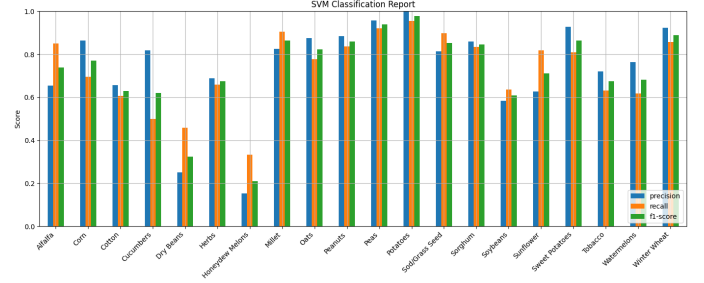
CNN Hard Multi



RF Hard Multi



LSTM Hard Multi



SVM Hard Multi

Fig. 15. Per-class accuracy metrics (precision, recall, F1) for the Hard Multi dataset across all models.